Unsupervised Segmentation of Retail Borrowers in the Loan Data 2007–2014 Dataset Using FAMD and K-Means Clustering

1. Abstract

This report presents an unsupervised learning analysis of the Loan Data 2007–2014 dataset with the aim to identify economically meaningful borrower segments in a portfolio. The analysis is based on a mixed data of borrowers and loan characteristics, including financial variables such as loan amount, income, interest rate, debt-to-income ratio, revolving credit utilization, and credit history depth, with categorical descriptors such as employment length, home ownership, verification status, and loan purpose. Since the data contains both quantitative and qualitative variables we utilized Factor Analysis of Mixed Data (FAMD) to reduce dimensionality and summarize the main structure of borrower heterogeneity, then applying k-means clustering. Although a formal cluster-selection benchmark indicated a broad two group structure, a four cluster result was chosen for interpretation, since it provided a richer and more economically useful segmentation of the portfolio. The results uncover distinct borrower profiles, which are: stressed high-rate borrowers, mainstream revolving-credit users, conservative small-balance borrowers, and affluent borrowers with deeper and more seasoned credit histories. Overall, the analysis shows that the portfolio is structured not by a single characteristic, but by the interaction of borrowing scale, repayment burden, revolving-credit stress, and credit maturity.

2. Statement of the problem / goal of the analysis and description of the dataset

The purpose of this analysis is to identify meaningful borrower clusters inside a retail lending portfolio using FAMD and k-means. It was important to understand the internal structure of the dataset whether borrowers can be grouped into distinct profiles based on their financial situation, loan characteristics, and credit history. So the goal is to see whether the portfolio is made up of clearly interpretable borrower types that differ in borrowing scale, repayment burden, use of revolving credit, and depth of credit history. The analysis uses the Loan Data 2007–2014 dataset, which contains loan-level and borrower-level information observed at origination. For the unsupervised part, the selected variables describe both the loan itself and the borrower's financial profile. These include loan amount, loan term, interest rate, installment, annual income, debt-to-income ratio, delinquencies in the previous two years, recent credit inquiries, number of open accounts, public derogatory records, revolving balance, revolving utilization, total number of accounts, employment length, home ownership, verification status, loan purpose, and initial listing status. The dataset also includes date variables that make it possible to construct a measure of credit-history length. Loan grade and loan status were kept only for interpreting the clusters afterward and were not used to create them. Prior to application of FAMD the data was cleaned and standardized, numeric fields stored as text were converted into proper numeric form, percentage signs and commas were removed where needed, and loan term was rewritten as a numeric number of months. Employment length was recoded into consistent categories, while the issue date and earliest credit-line date were converted from month-year strings into usable date variables. Furthemore this dates were converted into a usable credit length variable. Because some monetary variables were large, loan amount, installment, annual income, and revolving balance were log-transformed to dial them down for a more balanced observation. After keeping only observations with complete values in the active clustering variables, the final dataset contained 464,776 loans. This dataset is suited to our research since it combines continuous financial variables with categorical borrower characteristics. Which makes it possible to study borrower heterogeneity in a richer way than a simple rich-poor split. The aim of the analysis is therefore not just to divide the sample into groups, but to obtain segments that are interpretable from a financial point of view and useful for describing how the lending portfolio is actually structured.

3. Findings / key points

1. **The structure of the portfolio is driven by several financial characteristics at the same time, not by one variable alone.**
   The analysis shows that borrower differences can`t be explained by looking at income, loan size, or credit grade in isolation. What matters is the combination of factors such as how much the borrower is taking on, how heavy the repayment burden is, how intensively revolving credit is being used, and how deep or mature the credit history is. So the portfolio is organised around a broader financial profile not just around one dominant measure. This is an important result because it shows that borrower heterogeneity is more complex than a simple high-income versus low-income or safe versus risky distinction.
2. **A broad two-group structure is present in the data, but it is too simple to describe the portfolio in a useful way.**
   The formal cluster-selection benchmark pointed to a two-cluster solution, which suggests that at the highest level the portfolio can indeed be split into two broad groups. However, that solution is quite coarse for interpretation, since it merged together borrowers who differ in economically meaningful ways. The two-cluster result captures the existence of a general divide in the portfolio, but it does not explain the internal structure of that divide. For that reason, a more detailed segmentation was needed to produce a result that is useful from a financial point of view.
3. **The four-cluster solution reveals four borrower profiles that are distinct and economically interpretable.**
   The final segmentation identifies four groups that differ in a way that is meaningful for understanding the lending portfolio. One group consists of financially stretched borrowers who face higher interest rates, higher repayment pressure, and heavier revolving-credit use. A second group represents more typical mid-balance borrowers who are not the riskiest, but still rely actively on revolving credit. A third group is made up of more conservative borrowers who take smaller loans, carry lower repayment burden, and appear more cautious in their use of credit. Finally, the fourth group consists of more affluent and established borrowers, characterized by higher incomes, longer credit histories, and deeper account structures. Taken together, these clusters show that the portfolio is not simply divided into good and bad borrowers, but

into several different borrower types with different financial behaviours and credit profiles.

4. Analysis with commentary

To identify borrower segments, the analysis was carried out in two steps. First the mixed borrower dataset was reduced into a smaller number of latent components using Factor Analysis of Mixed Data (FAMD). FAMD was used since the data contained both numerical labels such as loan amount, income, interest rate, debt-to-income ratio, and use of revolving debt, and categorical information such as employment length, home ownership, verification status, and loan purpose. Second, we applied k-means clusterin to the first five FAMD components to find segmentations with similar characteristics. First benchmark revealed that there is a two way segmentation, but it was decided that solution was too general, since it mainly captured a high-level divide and didn`t describe the internal composition of the portfolio in enough detail. That's why a four-cluster solution was kept as the final specification. The new four cluster solution had a balanced divide as shown in **Table 1**, no cluster is negligible in size: the smallest group is 18.08% of the whole sample, while the largest is 29.9%. This reveals to us that there is a meaningful division in the portfolio.

Table 1. Cluster sizes in the final k = 4 solution

Cluster	Number of loans	Share of sample (%)
1	84,032	18.08
2	124,185	26.72
3	138,972	29.90
4	117,587	25.30

As seen in the Table 2 the clusters differ in several economical variables where cluster 2 stands out as the most stressed segment: it has the largest average loan amount, the highest average interest rate, the highest debt-to-income ratio, and the heaviest use of revolving debt. Cluster 4 describes individuals with the highest average income with the longest credit history and the deepest account structure, representing a more mature and established borrower profile. Cluster 1 is clearly different from both: it has the smallest average loan amount, the smallest installment, the lowest debt burden, and the lowest use of revolving debt, making it the most conservative borrower segment. Cluster 3 occupies a middle position, with moderate loan size and income but relatively active use of revolving debt, which makes it more representative of a mainstream retail borrowing profile.

Table 2. Numerical profile of the four borrower clusters

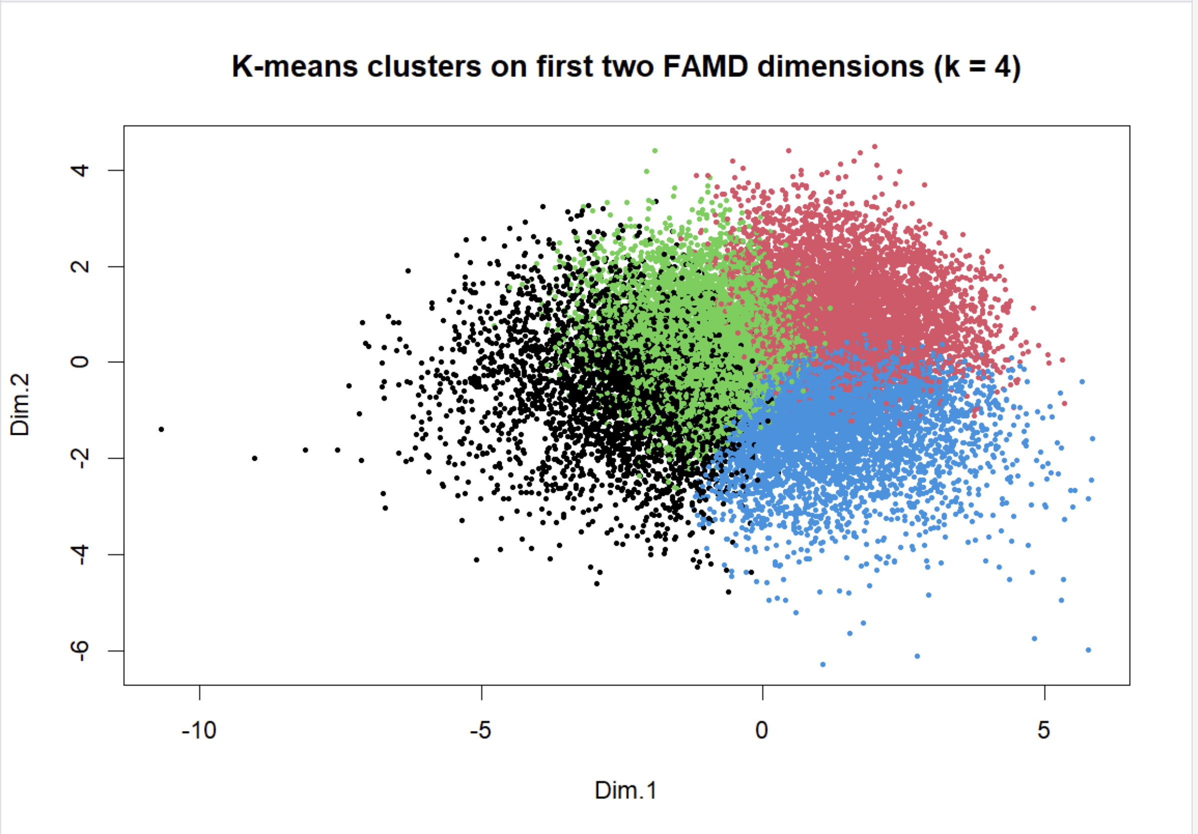
Cluster	Share (%)	Avg. loan amount	Avg. income	Avg. interest rate	Avg. DTI	Avg. revolving debt use	Avg. credit history (months)
1	18.08	6,017	51,259	13.44	14.21	40.14	174.0
2	26.72	21,257	82,238	17.67	18.88	65.83	197.0
3	29.90	10,840	56,947	12.70	16.84	62.26	151.1
4	25.30	17,032	98,763	11.39	18.07	50.28	241.1

Based on the results we can identify different styles of borrowing and credit use. Some borrowers appear financially stretched and expensive to lend to, some look like ordinary retail borrowers with active revolving-debt use, some appear more cautious and lower-intensity, and others are more affluent and established. This is what makes the four-cluster solution more useful than a simpler split, since it divides the portfolio into 4 distinct groups that differ in credit use. For clarity, the final segmentation can be summarized in the simplified form shown in **Table 3**.

Table 3. Simple interpretation of the four clusters

Cluster	Short label	Main idea
1	Conservative small-balance borrowers	Small loans, lower burden, more cautious credit use
2	Stressed high-rate borrowers	Large loans, highest rates, highest DTI, heaviest credit-card type debt use
3	Mainstream revolving-debt users	Mid-size loans, ordinary profile, but active use of revolving debt
4	Affluent seasoned borrowers	Higher income, longer credit history, deeper credit profile

The cluster graph provides a useful visual summary of this segmentation. When the observations are plotted on the first two FAMD dimensions and coloured according to the final k-means assignment, four groups can be observed, although some overlap remains visible. This is not a problem, since the final clustering was obtained using the first five FAMD dimensions rather than only the two dimensions shown in the figure. The graph should therefore be read as a simplified visual representation of the segmentation rather than as the full basis of the clustering itself.



5. Conclusion

This analysis set out to examine whether a retail lending portfolio can be meaningfully segmented using unsupervised learning methods applied to origination-time borrower and loan characteristics. The results show that it can. Rather than revealing a single simple divide, the portfolio appears to be structured by a combination of borrowing scale, repayment burden, use of revolving debt, and depth of credit history. So we can deduce that borrower heterogeneity is clearly multidimensional and cannot be explained by income, loan size, or loan grade alone. A broad two-group structure was present in the data, but that solution was too coarse to describe the portfolio in a useful way. The four-cluster solution provided a more informative segmentation and revealed four distinct borrower profiles: conservative small-balance borrowers, stressed high-rate borrowers, mainstream revolving-debt users, and affluent seasoned borrowers. Together, these groups suggest that the lending portfolio is made up of several different borrowing styles rather than a simple split between safe and risky clients.

From a practical point of view, this segmentation can be useful because it gives a lender a more detailed view of how different borrowers are positioned within the portfolio. The stressed high-rate segment will require closer monitoring, tighter underwriting, and more conservative exposure management, since it combines high pricing with heavier financial pressure and stronger dependence on revolving debt. The mainstream revolving-debt segment may be better suited to standard retail strategies, but it could also be monitored for signs of growing debt dependence or deteriorating repayment capacity. The conservative small-balance users appear to be less intensive in its use of credit and may be relevant for retention strategies, low-intensity product offers, or gradual relationship expansion. The affluent seasoned segment may be attractive for relationship deepening, tailored pricing, or cross-selling, since it combines stronger income and longer credit history with a more established credit profile. The results reveal to us that borrower segmentation can complement traditional credit assessment by showing how risk, leverage, and credit usage are distributed across different parts of the portfolio. A bank could use this kind of segmentation for portfolio monitoring, customer targeting, early-warning systems, pricing differentiation, and product design. It should not replace supervised risk modelling, but it can provide a useful descriptive layer that helps explain the internal composition of the portfolio before prediction is introduced.